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import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load the dataset
df = pd.read_csv('/content/SampleData.csv')

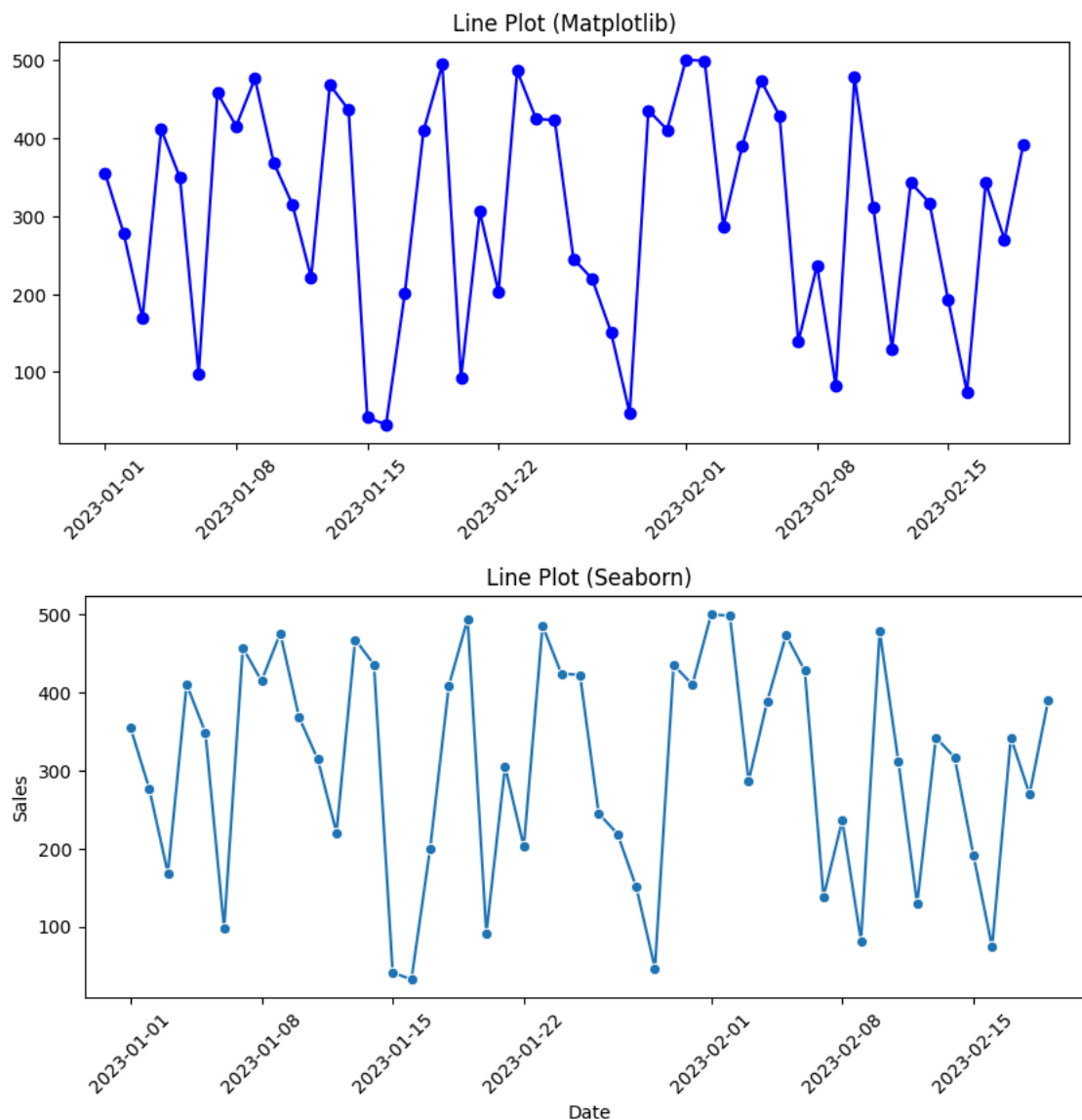
df['Date'] = pd.to_datetime(df['Date'])

df.head()

plot_data = df.head(50)
```

```
# Matplotlib Line Plot
plt.figure(figsize=(10, 4))
plt.plot(plot_data['Date'], plot_data['Sales'], color='blue', marker='o')
plt.title('Line Plot (Matplotlib)')
plt.xticks(rotation=45)
plt.show()

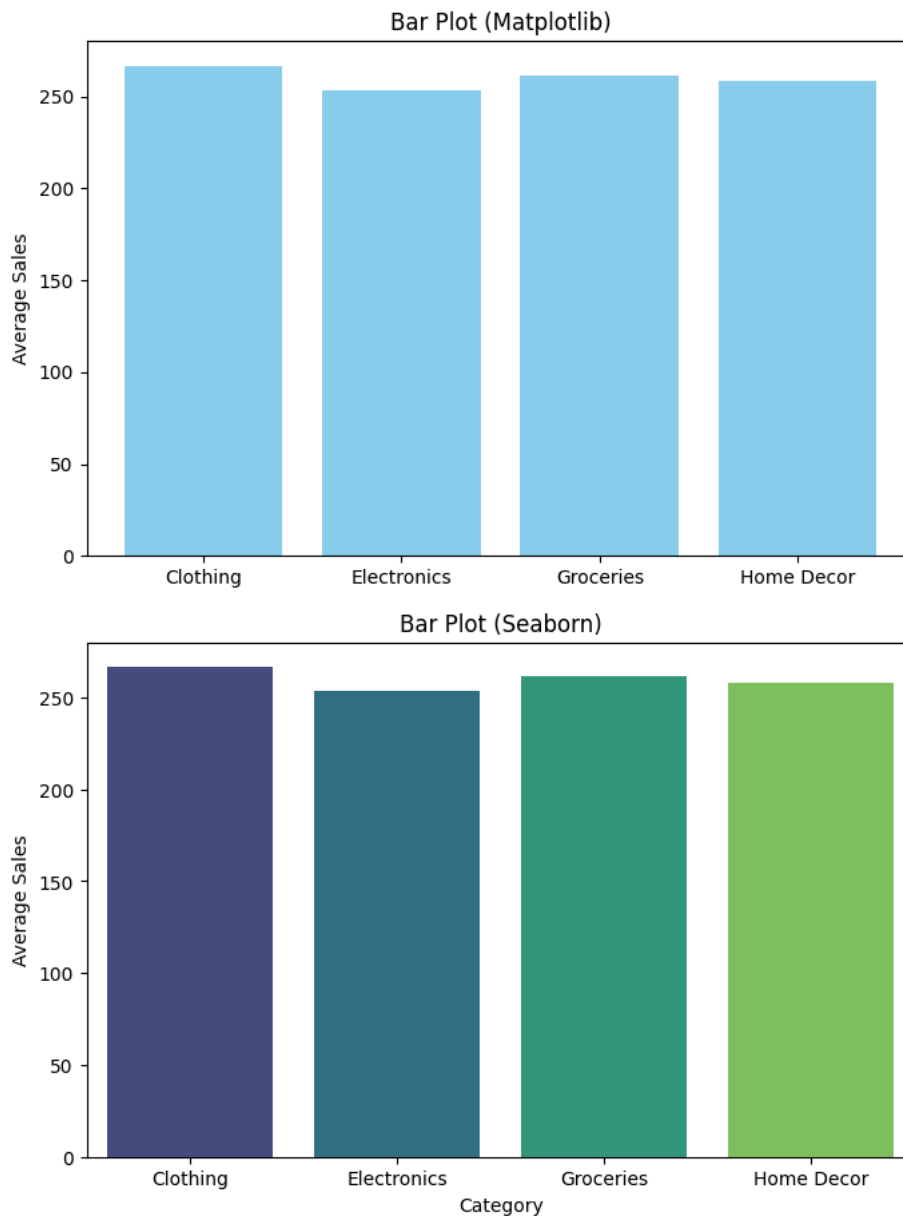
# Seaborn Line Plot
plt.figure(figsize=(10, 4))
sns.lineplot(data=plot_data, x='Date', y='Sales', marker='o')
plt.title('Line Plot (Seaborn)')
plt.xticks(rotation=45)
plt.show()
```



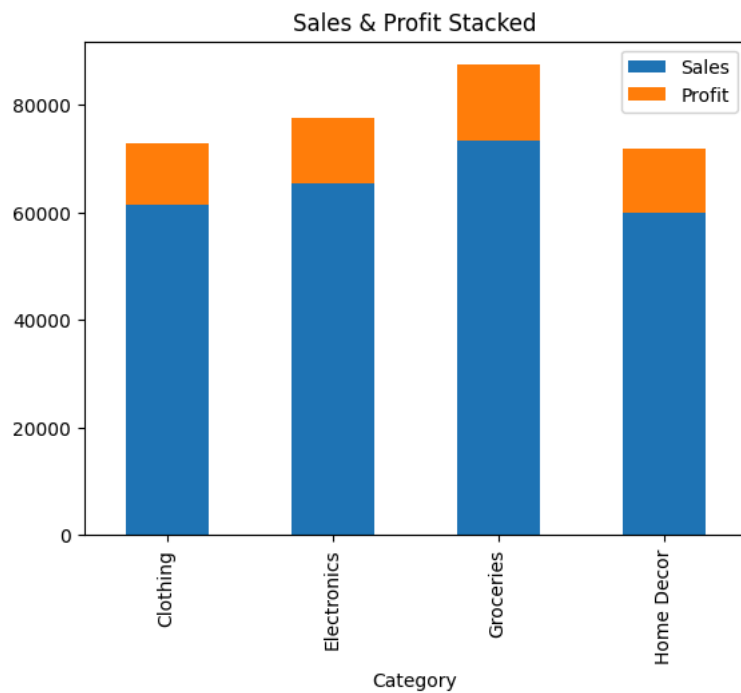
```
# Simple aggregation for average sales
avg_sales = df.groupby('Category')['Sales'].mean()
```

```
# Matplotlib Bar Plot
plt.figure(figsize=(8, 5))
plt.bar(avg_sales.index, avg_sales.values, color='skyblue')
plt.title('Bar Plot (Matplotlib)')
plt.ylabel('Average Sales')
plt.show()

# Seaborn Bar Plot
plt.figure(figsize=(8, 5))
sns.barplot(x=avg_sales.index, y=avg_sales.values, hue=avg_sales.index, palette='viridis')
plt.title('Bar Plot (Seaborn)')
plt.ylabel('Average Sales')
plt.show()
```

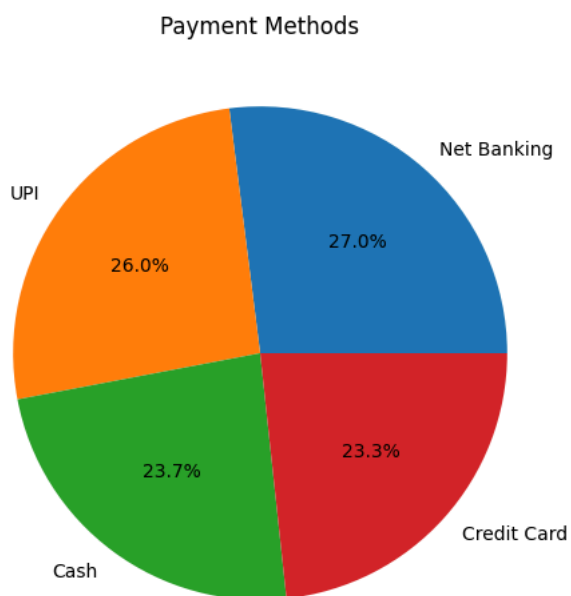


```
# Stacked Bar Chart
sbc = df.groupby('Category')[['Sales', 'Profit']].sum()
sbc.plot(kind='bar', stacked=True)
plt.title('Sales & Profit Stacked')
plt.show()
```



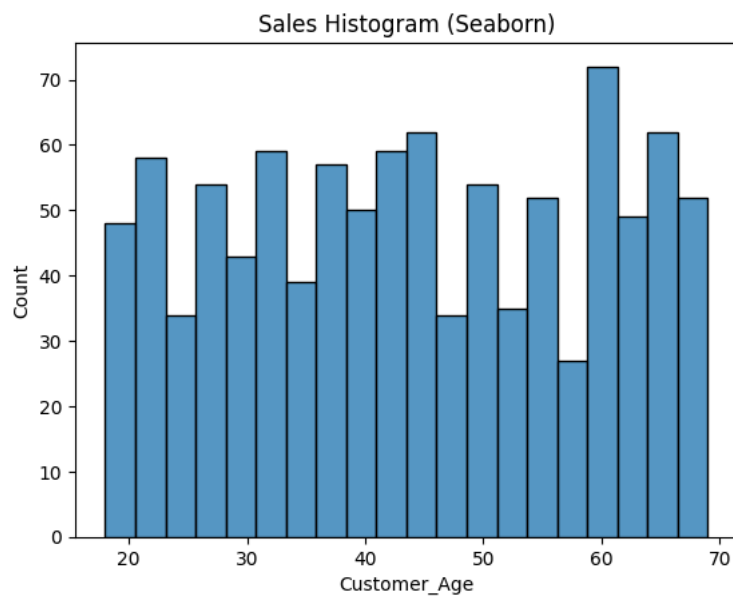
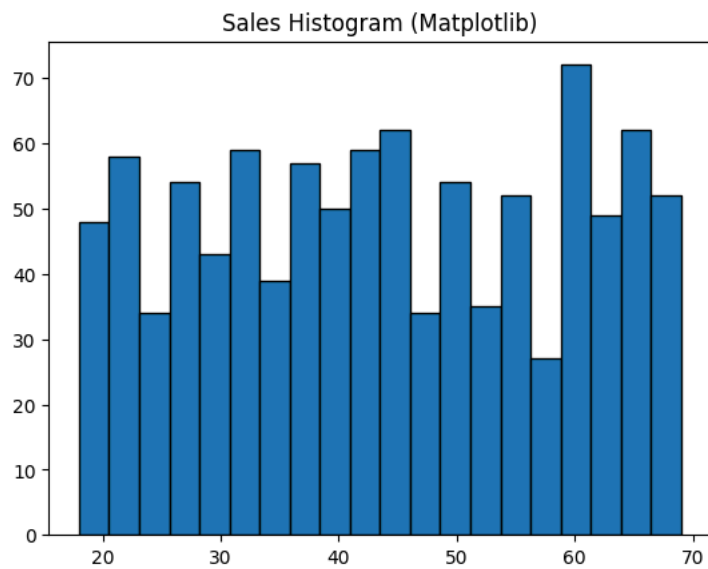
```
# Pie Chart
pay_counts = df['Payment_Method'].value_counts()

plt.figure(figsize=(6, 6))
plt.pie(pay_counts, labels=pay_counts.index, autopct='%1.1f%%')
plt.title('Payment Methods')
plt.show()
```

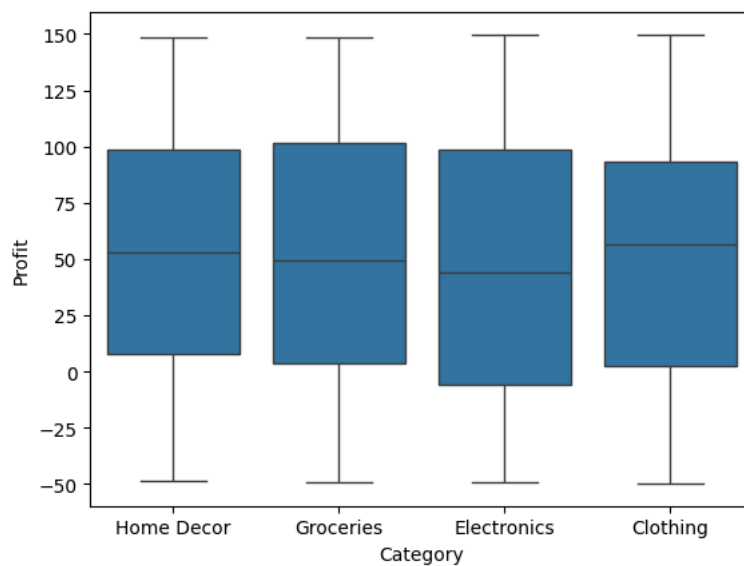


```
# Histogram (Matplotlib)
plt.title('Sales Histogram (Matplotlib)')
plt.hist(df['Customer_Age'], bins=20, edgecolor='black')
plt.show()

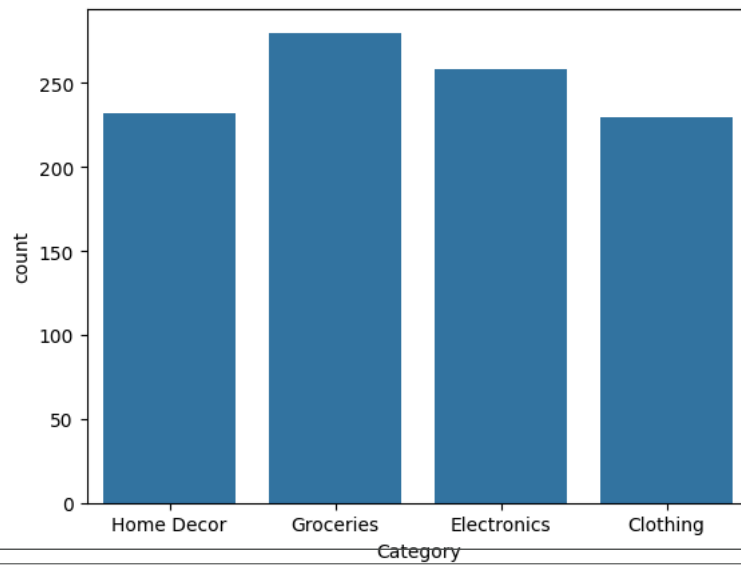
# Histogram (Seaborn)
plt.title('Sales Histogram (Seaborn)')
sns.histplot(df['Customer_Age'], bins=20, edgecolor='black')
plt.show()
```



```
# Box Plot
sns.boxplot(data=df, x='Category', y='Profit')
plt.show()
```

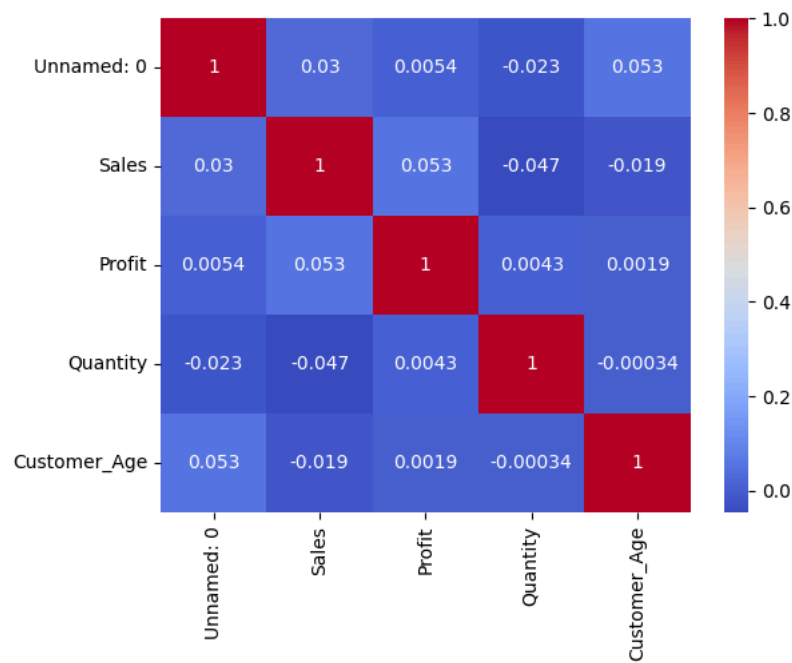


```
# Count Plot
sns.countplot(data=df, x='Category')
plt.show()
```



```
# Heatmap
x= df.corr(numeric_only=True)

sns.heatmap(x, annot=True, cmap='coolwarm')
plt.show()
```



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